

Conditional Probability



Basic probability rules

- 1 A bag contains 5 red marbles, 3 blue marbles, and 2 green marbles. Find the probability of drawing a blue marble.
 - 2 A fair six-sided die is rolled. Find the probability of rolling an even number or a 5.
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Two-way tables and conditional probability

- 3 A survey of 300 people found: 90 own a dog and a cat, 60 own only a dog, 40 own only a cat, and 110 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.
 - 4 Using the same survey data, find the probability a randomly selected person owns a dog, given that they own a cat.
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Independent vs. dependent events

- 5 Explain the difference between independent and dependent events, and classify the following: drawing two cards from a deck without replacement.
 - 6 A jar has 4 red and 6 blue chips. Find the probability of drawing two red chips in a row, without replacement.
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The multiplication rule for independent events

- 7 A fair coin is flipped and a fair six-sided die is rolled. Find the probability of getting heads and rolling a 4.
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Probability with replacement vs. without replacement

- 8 A deck of 10 cards numbered 1-10 has one card drawn, replaced, then a second card drawn. Find the probability both cards drawn are greater than 7.
 - 9 Using the same deck of 10 cards but WITHOUT replacement this time, find the probability both cards drawn are greater than 7.
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Interpreting conditional probability in real-world contexts

- 10** A medical test is 95% accurate for detecting a disease that affects 1% of the population. Explain, in words, why a positive test result does not necessarily mean a 95% chance of having the disease, referencing the low overall disease rate.
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The complement rule

- 11** A spinner has 8 equal sections numbered 1-8. Find the probability of NOT landing on a number greater than 6, using the complement rule.
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Probability trees for multi-stage events

- 12** A box has 3 red and 2 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.
- 13** Using the same box (3 red, 2 white, no replacement), find the probability of drawing two balls of the same color.
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Comparing theoretical and experimental probability

- 14** A fair coin is flipped 50 times and lands heads 32 times. Compare the experimental probability of heads to the theoretical probability, and explain a possible reason for the difference.
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Combining conditional probability with two-way tables

- 15** A two-way table of 400 students shows: 180 play an instrument and are in band; 20 play an instrument but are not in band; 40 don't play an instrument but are in band; 160 neither play an instrument nor are in band. Find the probability a student is in band, given that they play an instrument.
- 16** Using the same table (180 instrument+band, 20 instrument only, 40 band only, 160 neither, total 400), find the probability that a randomly selected student plays an instrument OR is in band (not both counted twice).
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A multi-part conditional probability problem

- 17 This question has three parts. A school has 600 students: 350 take Spanish, 200 take French, and 50 take both languages. (a) Find the probability a randomly selected student takes at least one of the two languages. (b) Find the probability a student takes Spanish, given that they take at least one language. (c) Find the probability a student takes neither language.
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Multiple choice — easy

- 18 A bag contains 9 red marbles, 12 blue marbles, and 4 green marbles. Find the probability of drawing a blue marble.
- a. $\frac{13}{25}$
 - b. $\frac{12}{25}$
 - c. $\frac{9}{25}$
 - d. $\frac{4}{25}$
- 19 A bag contains 11 orange marbles, 6 purple marbles, and 6 silver marbles. Find the probability of drawing a silver marble.
- a. $\frac{7}{23}$
 - b. $\frac{11}{23}$
 - c. $\frac{5}{23}$
 - d. $\frac{6}{23}$
- 20 A bag contains 8 orange marbles, 11 purple marbles, and 9 silver marbles. Find the probability of drawing a purple marble.
- a. $\frac{11}{28}$
 - b. $\frac{3}{7}$
 - c. $\frac{9}{28}$
 - d. $\frac{2}{7}$

- 21 A bag contains 4 red marbles, 9 blue marbles, and 6 green marbles. Find the probability of drawing a blue marble.
- a. $\frac{9}{19}$
 - b. $\frac{6}{19}$
 - c. $\frac{4}{19}$
 - d. $\frac{10}{19}$
- 22 A bag contains 4 black marbles, 4 white marbles, and 9 yellow marbles. Find the probability of drawing a black marble.
- a. $\frac{4}{17}$
 - b. $\frac{5}{17}$
 - c. $\frac{9}{17}$
 - d. $\frac{3}{17}$
- 23 A bag contains 8 black marbles, 2 white marbles, and 10 yellow marbles. Find the probability of drawing a black marble.
- a. $\frac{2}{5}$
 - b. $\frac{1}{2}$
 - c. $\frac{1}{10}$
 - d. $\frac{9}{20}$
- 24 A bag contains 9 red marbles, 10 blue marbles, and 12 green marbles. Find the probability of drawing a red marble.
- a. $\frac{12}{31}$
 - b. $\frac{10}{31}$
 - c. $\frac{9}{31}$
 - d. $\frac{8}{31}$

- 25 A bag contains 5 red marbles, 2 blue marbles, and 8 green marbles. Find the probability of drawing a green marble.
- a. $\frac{2}{15}$
 - b. $\frac{3}{5}$
 - c. $\frac{1}{3}$
 - d. $\frac{8}{15}$
- 26 A bag contains 6 black marbles, 8 white marbles, and 5 yellow marbles. Find the probability of drawing a black marble.
- a. $\frac{7}{19}$
 - b. $\frac{8}{19}$
 - c. $\frac{6}{19}$
 - d. $\frac{5}{19}$
- 27 A bag contains 5 orange marbles, 10 purple marbles, and 8 silver marbles. Find the probability of drawing a purple marble.
- a. $\frac{8}{23}$
 - b. $\frac{5}{23}$
 - c. $\frac{10}{23}$
 - d. $\frac{11}{23}$
- 28 A bag contains 5 black marbles, 9 white marbles, and 5 yellow marbles. Find the probability of drawing a white marble.
- a. $\frac{5}{19}$
 - b. $\frac{10}{19}$
 - c. $\frac{9}{19}$
 - d. $\frac{8}{19}$

- 29** A bag contains 7 black marbles, 8 white marbles, and 5 yellow marbles. Find the probability of drawing a black marble.
- a. $\frac{3}{10}$
 - b. $\frac{2}{5}$
 - c. $\frac{1}{4}$
 - d. $\frac{7}{20}$
- 30** A bag contains 7 black marbles, 12 white marbles, and 9 yellow marbles. Find the probability of drawing a yellow marble.
- a. $\frac{9}{28}$
 - b. $\frac{1}{4}$
 - c. $\frac{5}{14}$
 - d. $\frac{3}{7}$
- 31** A bag contains 8 orange marbles, 11 purple marbles, and 7 silver marbles. Find the probability of drawing a purple marble.
- a. $\frac{6}{13}$
 - b. $\frac{4}{13}$
 - c. $\frac{11}{26}$
 - d. $\frac{7}{26}$
- 32** A bag contains 5 orange marbles, 9 purple marbles, and 10 silver marbles. Find the probability of drawing a purple marble.
- a. $\frac{3}{8}$
 - b. $\frac{1}{3}$
 - c. $\frac{5}{12}$
 - d. $\frac{5}{24}$

Multiple choice — medium

- 33** A survey of 83 people found: 47 own a car and a bicycle, 5 own only a car, 5 own only a bicycle, and 26 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.
- a. $\frac{47}{83}$
 - b. 1
 - c. $\frac{25}{26}$
 - d. $\frac{47}{52}$
- 34** A survey of 73 people found: 21 own a dog and a cat, 9 own only a dog, 20 own only a cat, and 23 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.
- a. $\frac{21}{41}$
 - b. $\frac{21}{73}$
 - c. $\frac{7}{10}$
 - d. $\frac{41}{30}$
- 35** A survey of 83 people found: 21 own a car and a bicycle, 16 own only a car, 26 own only a bicycle, and 20 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.
- a. $\frac{21}{83}$
 - b. $\frac{21}{47}$
 - c. $\frac{21}{37}$
 - d. $\frac{47}{37}$

- 36** A survey of 71 people found: 27 own a piano and a guitar, 30 own only a piano, 8 own only a guitar, and 6 own neither. Find the probability a randomly selected person owns a guitar, given that they own a piano.
- a. $\frac{27}{71}$
 - b. $\frac{9}{19}$
 - c. $\frac{27}{35}$
 - d. $\frac{35}{57}$
- 37** A survey of 98 people found: 46 own a smartphone and a smartwatch, 11 own only a smartphone, 26 own only a smartwatch, and 15 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.
- a. $\frac{24}{19}$
 - b. $\frac{23}{36}$
 - c. $\frac{23}{49}$
 - d. $\frac{46}{57}$
- 38** A survey of 134 people found: 32 own a piano and a guitar, 35 own only a piano, 36 own only a guitar, and 31 own neither. Find the probability a randomly selected person owns a guitar, given that they own a piano.
- a. $\frac{8}{17}$
 - b. $\frac{68}{67}$
 - c. $\frac{16}{67}$
 - d. $\frac{32}{67}$

- 39 A survey of 92 people found: 22 own a smartphone and a smartwatch, 14 own only a smartphone, 22 own only a smartwatch, and 34 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.
- a. $\frac{1}{2}$
 - b. $\frac{11}{9}$
 - c. $\frac{11}{18}$
 - d. $\frac{11}{46}$
- 40 A survey of 80 people found: 20 own a car and a bicycle, 29 own only a car, 13 own only a bicycle, and 18 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.
- a. $\frac{33}{49}$
 - b. $\frac{20}{49}$
 - c. $\frac{1}{4}$
 - d. $\frac{20}{33}$
- 41 A survey of 78 people found: 43 own a dog and a cat, 23 own only a dog, 7 own only a cat, and 5 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.
- a. $\frac{43}{66}$
 - b. $\frac{43}{78}$
 - c. $\frac{43}{50}$
 - d. $\frac{25}{33}$
- 42 A survey of 53 people found: 15 own a dog and a cat, 22 own only a dog, 6 own only a cat, and 10 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.
- a. $\frac{15}{53}$
 - b. $\frac{5}{7}$
 - c. $\frac{15}{37}$
 - d. $\frac{21}{37}$

- 43 A survey of 97 people found: 59 own a car and a bicycle, 13 own only a car, 12 own only a bicycle, and 13 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.
- a. $\frac{71}{72}$
 - b. $\frac{59}{71}$
 - c. $\frac{59}{97}$
 - d. $\frac{59}{72}$
- 44 A survey of 75 people found: 20 own a dog and a cat, 22 own only a dog, 7 own only a cat, and 26 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.
- a. $\frac{9}{14}$
 - b. $\frac{20}{27}$
 - c. $\frac{10}{21}$
 - d. $\frac{4}{15}$
- 45 A survey of 67 people found: 15 own a piano and a guitar, 36 own only a piano, 6 own only a guitar, and 10 own neither. Find the probability a randomly selected person owns a guitar, given that they own a piano.
- a. $\frac{5}{7}$
 - b. $\frac{7}{17}$
 - c. $\frac{5}{17}$
 - d. $\frac{15}{67}$

- 46 A survey of 107 people found: 35 own a smartphone and a smartwatch, 23 own only a smartphone, 33 own only a smartwatch, and 16 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.
- a. $\frac{35}{58}$
 - b. $\frac{35}{68}$
 - c. $\frac{34}{29}$
 - d. $\frac{35}{107}$
- 47 A survey of 84 people found: 20 own a smartphone and a smartwatch, 8 own only a smartphone, 19 own only a smartwatch, and 37 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.
- a. $\frac{39}{28}$
 - b. $\frac{5}{7}$
 - c. $\frac{5}{21}$
 - d. $\frac{20}{39}$
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Multiple choice — hard

- 48 Two independent events occur: a spinner (5 equal sections) lands on section 1, and a fair six-sided die shows an even number. Find the probability both events occur.
- a. $\frac{1}{10}$
 - b. $\frac{1}{5}$
 - c. $\frac{1}{20}$
 - d. $\frac{7}{10}$

- 49 Two independent events occur: a spinner (5 equal sections) lands on section 1, and a card drawn from a standard deck is a heart. Find the probability both events occur.
- a. $\frac{1}{20}$
 - b. $\frac{3}{20}$
 - c. $\frac{1}{10}$
 - d. $\frac{9}{20}$
- 50 Two independent events occur: a card drawn from a standard deck is a heart, and a fair coin lands heads. Find the probability both events occur.
- a. $\frac{1}{4}$
 - b. $\frac{9}{40}$
 - c. $\frac{1}{8}$
 - d. $\frac{3}{4}$
- 51 Two independent events occur: a fair six-sided die shows an even number, and a spinner (5 equal sections) lands on section 1. Find the probability both events occur.
- a. $\frac{3}{20}$
 - b. $\frac{7}{10}$
 - c. $\frac{1}{10}$
 - d. $\frac{1}{5}$
- 52 Two independent events occur: a fair coin lands heads, and a fair six-sided die shows an even number. Find the probability both events occur.
- a. $\frac{1}{2}$
 - b. 1
 - c. $\frac{7}{20}$
 - d. $\frac{1}{4}$

- 53 A box has 5 red and 4 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.
- a. $\frac{5}{18}$
 - b. $\frac{17}{45}$
 - c. $\frac{5}{9}$
 - d. $\frac{59}{180}$
- 54 A box has 4 red and 7 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.
- a. $\frac{28}{55}$
 - b. $\frac{7}{11}$
 - c. $\frac{67}{220}$
 - d. $\frac{14}{55}$
- 55 A box has 6 red and 6 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.
- a. $\frac{6}{11}$
 - b. $\frac{3}{11}$
 - c. $\frac{71}{220}$
 - d. $\frac{1}{2}$
- 56 A box has 6 red and 3 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.
- a. $\frac{1}{3}$
 - b. $\frac{1}{4}$
 - c. $\frac{3}{10}$
 - d. $\frac{1}{2}$

- 57 A box has 8 red and 7 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.
- a. $\frac{7}{15}$
 - b. $\frac{19}{60}$
 - c. $\frac{4}{15}$
 - d. $\frac{8}{15}$
- 58 A spinner has 15 equal sections numbered 1 to 15. Find the probability of NOT landing on one of the 5 sections numbered 1 through 5, using the complement rule.
- a. $\frac{2}{5}$
 - b. $\frac{4}{15}$
 - c. $\frac{1}{3}$
 - d. $\frac{2}{3}$
- 59 A spinner has 15 equal sections numbered 1 to 15. Find the probability of NOT landing on one of the 2 sections numbered 1 through 2, using the complement rule.
- a. $\frac{13}{15}$
 - b. $\frac{2}{15}$
 - c. $\frac{1}{5}$
 - d. $\frac{1}{15}$
- 60 A spinner has 7 equal sections numbered 1 to 7. Find the probability of NOT landing on one of the 2 sections numbered 1 through 2, using the complement rule.
- a. $\frac{3}{7}$
 - b. $\frac{5}{7}$
 - c. $\frac{1}{7}$
 - d. $\frac{2}{7}$

61 A spinner has 9 equal sections numbered 1 to 9. Find the probability of NOT landing on one of the 8 sections numbered 1 through 8, using the complement rule.

a. $\frac{1}{3}$

b. $\frac{1}{9}$

c. $\frac{8}{9}$

d. $\frac{7}{9}$

62 A spinner has 13 equal sections numbered 1 to 13. Find the probability of NOT landing on one of the 10 sections numbered 1 through 10, using the complement rule.

a. $\frac{10}{13}$

b. $\frac{3}{13}$

c. $\frac{9}{13}$

d. $\frac{11}{13}$